Episode: Authentication with LDAP

- LDAP Authentication
  - What is LDAP?
    - *Lightweight Directory Access Protocol*
    - Is like a digital "phonebook" for resources
      - https://www.openldap.org/
      - LDAP gives you the ability to query Directory Services
        - This is basically a network-based database of network resources
          - Files
          - Printers
          - Servers
          - Users
            - Is Windows Active Directory and LDAP the same thing?
              - No, but AD uses LDAP
  - How is it used for Authentication?
    - *ldapAuthProcess app*
      1. Connection Establishment:
      - The client, which could be a user or an application, initiates a connection to the LDAP server using the LDAP protocol (usually over port 389 or over SSL on port 636).
      2. Binding:
      - The client sends a "bind" request to the LDAP server. This request contains the user's distinguished name (DN) and a password. This can be an anonymous bind, a simple bind (username and password), or other supported authentication methods like SASL (Simple Authentication and Security Layer).
      3. Server Validation:
      - The LDAP server checks the provided credentials against the directory information. If the credentials are correct, the server responds with a success message. If not, it returns an error indicating authentication failure.
      4. Search (Optional):
      - Upon successful binding, the client may perform a search to retrieve additional information from the directory, such as user attributes or group memberships.
      5. Authentication Response:
      - If the server validates the credentials, the client receives confirmation and is granted access to the directory information or any services the user is authorized to access.
      6. Session Establishment:
      - After successful authentication, the user/client can establish a session to access the requested resources or services.